

What GraphQA Measures at $n=40$, and What Structural Primers Add to It

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Abstract

Prepending a *primer* of factual sentences about a graph’s structure to its text encoding is a natural way to help a language model answer graph-reasoning questions. We evaluate primers on six GraphQA tasks over 40-node Erdős–Rényi graphs with four Qwen3 arms, against four controls applied jointly: a graph-blind solver refit at the evaluated graph size, a length-matched content-free filler, a minimal-length content control, and the majority-class baseline of each task. Reading accuracy against the class prior rather than against zero reorganises the benchmark: two of the six tasks have a prior of 1.000 at this graph size, and on a third both plain arms score below the prior at every density. Within what remains, primers act through two mechanisms. A primer that states the requested quantity is a second route to the answer, helping a model at floor and harming one at ceiling (cross-fitted interaction -11.0 points, 95% CI $[-17.1, -5.7]$); the relation holds on `node_degree` alone, which truncates on under 2% of generations, so it is not an artifact of censoring. Added text is an active treatment whose cost is the only effect that replicates across model sizes. Two primers that do not state the answer beat both controls, and a fixed-mean-degree sweep over $n \in \{20, 40, 80, 160\}$ reproduces one of them at $+6.2$ points ($n=3,191$, $p < 10^{-4}$).

1 Introduction

Fatemi et al. (2023) show that a language model’s accuracy on graph-reasoning questions varies with the text encoding when the graph and question are held fixed. We ask whether prepending a short *primer* of derived, node-level facts to that encoding improves the answers.

Four confounds complicate any such measurement. A primer may supply the answer directly, so the model need not read the graph (answer leakage). A primer changes prompt length, and length alone

affects accuracy (Levy et al., 2024; Liu et al., 2024). A model near ceiling leaves no room for an effect. And a task whose answers are unbalanced can be scored well by a constant response, so accuracy overstates competence.

The fourth is usually left implicit, and we make it the starting point: Section 3 measures what each of the six tasks can register at $n=40$ before any primer is applied. This is not a preliminary — it changes which results are interpretable. Sections 4–6 then address three questions. **RQ1**: when a primer helps, is it because it states the answer? **RQ2**: does added text affect accuracy independently of its content? **RQ3**: does any primer that does not state the answer beat a length-matched control?

Closest to a primer are the prompt-level interventions of Wang et al. (2023); Perozzi et al. (2024) instead learn a continuous encoding. Dziri et al. (2023) argue that transformers solve compositional tasks by linearised subgraph matching, which raises exactly the question a primer poses, and Geirhos et al. (2020) frame it as shortcut learning; our graph-blind solver operationalises that concern. Irrelevant context degrades reasoning (Shi et al., 2023), which our `filler` condition measures directly.

2 Method and Setup

Primers. Seven conditions: `none`; `degree`, `clustering`, `rwse` and `components`, stating each node’s degree, local clustering coefficient, random-walk return probabilities (Dwivedi et al., 2022) and the graph’s component count; `all`; and `filler`, content-free text matched to `clustering`’s length to within 7.8%. All seven pass through one renderer, so conditions differ in content, never in format. `components` is the minimal-length content control: it adds 9 tokens where `filler` adds 457.

Controls. A *graph-blind solver* parses the rendered primer text — never the graph — and answers from it alone, giving the score a primer-only program reaches. It is refit at $n=40$, because bars fitted on the published 5–19 node split do not transfer. Fitted rules use disjoint fit and score sets. A condition is a *route* for a task when its refit bar exceeds the control’s; on this corpus the split is unambiguous, the largest sub-threshold gain being 0.011 and the smallest supra-threshold 0.066.

Baselines. For every (task, density) we report the *majority-class baseline*: the share of the modal gold answer, which is what a constant response scores. Accuracy is read against this, not against zero.

Sweep. Erdős–Rényi graphs (Erdős and Rényi, 1959) with $n=40$ at $p \in \{.10, .20, .35, .50\}$, 100 graphs per (task, density) cell, seven conditions, six tasks: 16,800 prompts per arm. Arms are Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025), each with native thinking mode off and on. A high-density extension covers *node_degree* and *edge_existence* at $p \in \{.65, .75, .85\}$. Decoding is greedy, with 2,048 new tokens for plain arms and 8,192 for thinking arms. Two further corpora appear in Section 3: the published 5–19 node split for *cycle_check* and *edge_count*, and a fixed-mean-degree sweep in Section 6 that holds mean degree at ≈ 8 or ≈ 16 while n ranges over $\{20, 40, 80, 160\}$.

Metrics and tests. Exact match throughout; set-F1 on *connected_nodes* runs 0.96–1.00 in every cell and does not discriminate. Generations that reach the output budget are excluded pairwise; rates are under 6.5% except *edge_count*, which reaches 22.4% on qwen3-1.7b and 62% or more on the thinking arms. Per-cell tests are exact McNemar with Benjamini–Hochberg within each (arm, task) family at $q=0.05$; pooled effects use a cluster-robust paired permutation test and a cluster bootstrap CI, both with 10,000 draws. Nulls carry a minimum detectable effect.

3 What the Six Tasks Measure

Table 1 gives the majority-class baseline against measured accuracy under *none*. Four of the six tasks are compromised at this graph size, in four different ways.

Two tasks have a prior of 1.000. Every one of the 400 *cycle_check* graphs contains a cycle, at every density: at $n=40$ an Erdős–Rényi graph is far past the threshold for acyclicity, so the task is answered by a constant. *node_count*’s gold answer is 40 throughout. Together they are a third of the benchmark, and no primer effect on them is evidence about graph reading.

***node_count* fails below its own prior, through one error.** qwen3-1.7b scores 0.020, 0.000, 0.280 and 0.960 as density rises, against a prior of 1.000. Every wrong answer is “39”: the rate is 98%, 100%, 72% and 4% across the four densities. The encoder’s preamble names the nodes as “0, ..., 39”, and the model returns the range endpoint instead of the count. This is not an artifact of nodes missing from the encoding body: at $p=.20$ the body lists all 40 nodes for every graph, and the model still answers “39” on 100 of 100. We report the pattern without explaining why density removes it.

***edge_count* is below its prior everywhere, and the failure is arithmetic rather than perceptual.** Both plain arms score at or under the prior at all four densities. Exact match hides the shape of the error: median relative error for qwen3-4b is 8.1%, 9.7%, 12.6% and 17.2%, with *no* truncated generations, so the model is estimating, not failing to read. The thinking arms are near-exact where they terminate (0.960 for qwen3-1.7b-think at $p=.10$; 0.975, 1.000 and 0.875 for qwen3-4b-think) but truncate on 50–100% of instances. Counting edges is a serial enumeration that fits the budget only on sparse graphs; without it, models return a plausible magnitude.

***edge_existence* accuracy tracks the prior, for a reason that is not competence.** qwen3-1.7b scores within 9 points of the class prior at all seven densities. Decomposing into hits and false alarms shows why: its hit rate is 1.000 at every density, while its false-alarm rate climbs $0.122 \rightarrow 0.329 \rightarrow 0.493 \rightarrow 0.885 \rightarrow 0.941 \rightarrow 0.958 \rightarrow 0.933$. The model is not answering the prior — it is saying “yes” ever more freely, and “always yes” happens to *be* the majority strategy at high density. qwen3-4b holds false alarms between 0.044 and 0.375 and clears the prior by 6 to 38 points. With hits at ceiling, sensitivity and criterion are not separately identifiable (Macmillan

Task		0.1	0.2	0.35	0.5	0.65	0.75	0.85
node_count	prior	1.000	1.000	1.000	1.000	—	—	—
	1.7b	0.020	0.000	0.280	0.960	—	—	—
	4b	0.920	0.690	1.000	1.000	—	—	—
cycle_check	prior	1.000	1.000	1.000	1.000	—	—	—
	1.7b	0.988	1.000	1.000	1.000	—	—	—
	4b	0.990	1.000	1.000	1.000	—	—	—
edge_existence	prior	0.900	0.820	0.690	0.520	0.660	0.760	0.850
	1.7b	0.890	0.730	0.660	0.540	0.680	0.770	0.860
	4b	0.960	0.930	0.830	0.900	0.870	0.870	0.960
node_degree	prior	0.210	0.170	0.170	0.180	0.170	0.160	0.200
	1.7b	0.910	0.800	0.410	0.300	0.081	0.160	0.050
	4b	1.000	0.990	0.990	0.990	0.820	0.500	0.330
edge_count	prior	0.080	0.060	0.060	0.050	—	—	—
	1.7b	0.043	0.022	0.000	0.000	—	—	—
	4b	0.050	0.020	0.010	0.000	—	—	—
connected_nodes	prior	0.020	0.010	0.010	0.010	—	—	—
	1.7b	0.920	0.910	0.590	0.480	—	—	—
	4b	0.930	0.880	0.840	0.790	—	—	—

Table 1: Majority-class baseline (*prior*: the share of the modal gold answer, i.e. what a graph-blind constant answer scores) against measured exact-match accuracy under *none*, by edge density. Two of six tasks have a prior of 1.000 at $n=40$. Truncated generations excluded.

and Creelman, 2005; Hautus, 1995); accuracy on this task should not be reported alone.

Only node_degree and connected_nodes give graded signal at both sizes, and even node_degree falls below its prior for qwen3-1.7b at $p \geq .65$ (0.081, 0.160, 0.050 against 0.170, 0.160, 0.200).

On the published 5–19 node split the two degenerate tasks recover, and remain hard. There cycle_check has a prior of 83.2% (416 of 500 graphs cyclic). qwen3-0.6b scores 78.8% under *none* — below the prior — and primers raise it only to it (83.4% for components, $p < 10^{-3}$). edge_count is tractable on small graphs (23.4% for qwen3-1.7b, 50.0% for qwen3-8b, against a 3.4% prior) and collapses to $\leq 5\%$ at $n=40$. So neither task is inherently uninformative; both are saturated by the choice of $n=40$.

4 Answer-Stating Primers Substitute a Route

A *cell* is an (arm, task, density, condition) combination with at least ten usable pairs. Route cells gain where accuracy under *none* is low and lose where it is high; no-route cells show no such relation (Figure 1). Because baseline and effect share sampling noise, we estimate the route $\times$ baseline interaction on cross-fitted halves of each cell’s graphs: -11.0 points per unit of baseline, 95% CI $[-17.1, -5.7]$.

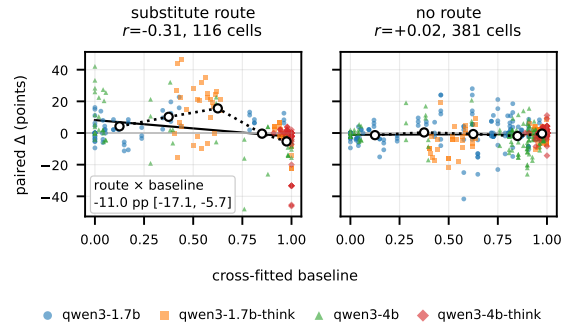


Figure 1: Cross-fitted paired effect against cross-fitted baseline accuracy, for cells where a primer offers a substitute route to the answer (left) and where it does not (right). Baseline and effect come from disjoint halves of each cell’s graphs.

A stated answer helps a model that cannot compute the answer and hurts one that already can.

Two checks bound the claim. First, the relation is *not graded*: within the 0.10–0.90 baseline window it disappears, so the data support a floor/ceiling contrast rather than a smooth dependence on competence. Second, it is not a censoring artifact. The pooled relation covers two tasks, one of which truncates heavily, so we partition it: over all 232 route cells $r = -0.308$ with slope -10.5 $[-16.6, -5.4]$; over the 168 node_degree cells alone $r = -0.235$ ($p=0.0022$) with slope -9.1 $[-21.6, -1.8]$; over the 64 edge_count cells $r = -0.664$ with slope -26.5 . node_degree truncates on 0–1.6% of generations and carries the

relation on its own.

Manipulating difficulty directly reproduces the contrast: for qwen3-1.7b-think on node_degree, raising density from $p=.10$ to $.85$ drives accuracy under none down and degree’s benefit up monotonically, from $+3.1$ to $+34.2$ points.

5 Length Is an Active Treatment

Every effect against none decomposes exactly into a length cost and a content gain,

$$\underbrace{c - n}_{\text{net}} = \underbrace{f - n}_{\text{length}} + \underbrace{c - f}_{\text{content}},$$

with n, f, c the accuracies under none, filler and the condition. Over the 90 cells with an uncontaminated primer and a baseline between 0.30 and 0.90, and correcting across all of them, the content gain is significant and positive in 29 cells against 2 negative, the length cost is significant and negative in 21, and the *net* survives in only 11. Content helps, length hurts, and they largely cancel.

filler states nothing about the graph, yet significantly reduces accuracy on connected_nodes and edge_existence in both plain arms — and across the four arms these are the only non-degenerate effects that replicate at both model sizes. The cost is not confined to this corpus: on the published split, rwse, the longest primer, costs qwen3-8b 13.3 points of edge_count accuracy (36.7% against 50.0%, $p < 10^{-3}$) and qwen3-1.7b 4.6 points ($p=0.036$).

A primer can also remove an answer rather than change it. Under degree, qwen3-1.7b-think fails to terminate on 19.2% of generations against 15.8% under none ($p=2.4 \times 10^{-6}$); outside edge_count the rates are 6.9% against 2.6% ($p=9.2 \times 10^{-11}$), 130 new failures attributable to the primer.

6 What Survives

On the main sweep, exactly two primers that do not state the answer beat both none and filler under a global correction. One is clustering on node_count, which Section 3 disqualifies. The other is components on connected_nodes for qwen3-4b: $+9.2$ points over none and $+21.8$ over filler ($n=400$, $p \approx 10^{-4}$ for both). Because components adds nine tokens, it is the only effect here that is essentially free of length

n	$\bar{d}$	qwen3-1.7b			qwen3-8b	
		none	clust.	Δ	none	Δ
20	8	0.912	0.865	-4.8^*	0.998	$+0.2$
40	8	0.728	0.770	$+4.2$	0.995	$+0.0$
80	8	0.645	0.750	$+10.5^*$	0.965	-0.8
160	8	0.624	0.693	$+6.9^*$	0.897	-2.8
20	16	0.575	0.740	$+16.5^*$	0.985	-0.2
40	16	0.398	0.476	$+7.8^*$	0.975	$+1.0$
80	16	0.225	0.323	$+9.8^*$	0.895	-1.8
160	16	0.224	0.211	-1.3	0.802	-4.0
pooled		0.542	0.604	$+6.2$	$n=3191$	

Table 2: Fixed-mean-degree sweep: mean degree $\bar{d}$ held at ≈ 8 or ≈ 16 while n varies, so density falls as the prompt lengthens. Δ is clustering against none in points. *: exact McNemar surviving Benjamini-Hochberg at $q=0.05$ across the eight cells. qwen3-8b is shown as a ceiling control. This corpus has no filler arm, so Δ is a net effect.

cost, and the gap to filler is mostly filler’s own damage.

clustering on node_degree is the effect a dedicated corpus was built to test. Pooled over $p=.35$ and $.50$ with 400 graphs per density it gains $+5.9$ points over none (95% CI $[2.1, 9.8]$), with an independent-seed replication at $+5.5$. Both the cell and the density window were chosen after the main sweep — where the same comparison gives $+3.0$, $p=0.26$ — so the replication, not the original estimate, is what makes it reportable.

A third corpus, built to separate graph size from answer magnitude, supports it on a design that played no part in that choice (Table 2). Holding mean degree at ≈ 8 or ≈ 16 while n ranges over $\{20, 40, 80, 160\}$, clustering gains $+6.2$ points over none pooled ($n=3,191$, 504 helped against 306 hurt, $p < 10^{-4}$), significant in six of eight cells after correction, and larger where the answer is larger ($+8.2$ at mean degree ≈ 16 against $+4.2$ at ≈ 8). Two caveats travel with it: the corpus has no filler arm, so this is a net effect and not evidence that content beats length; and qwen3-8b on the same grid sits at 0.998–0.802 and shows nothing, which is Section 4’s ceiling result rather than a contradiction.

The same grid dissociates two drivers of difficulty that density confounds. Holding mean degree fixed, growing n from 20 to 160 costs qwen3-1.7b 29 points and saturates after $n=80$; holding n fixed, doubling mean degree costs 33 to 42 points at every size. The magnitude of the answer matters more than the size of the graph.

Limitations

The main sweep is two sizes of one model family; the published-split and fixed-degree corpora add arms but were collected earlier and not pre-registered. All graphs are Erdős–Rényi, in which local clustering is nearly uniform, so clustering carries little node-specific information — a likely reason its effect is small. The route split depends on a threshold and on the $n=40$ re-fit. `filler` matches `clustering`’s length but is much shorter than `rwse` and `all`, so the length control is weakest exactly where length is largest. `clustering`’s window was chosen post hoc. Every prompt uses one encoding, one template and one primer position, with greedy decoding. Several cells rest on 400 or fewer instances. We have no mechanism for the `clustering` effect: rewiring that drives mean clustering from 0.01 to 0.73 leaves the benefit flat, and a density-prior account predicts a signed error shift of up to 17 where the observed shift stays within 0.3 of zero. Three further conditions would settle the open questions — a shuffled-value placebo stricter than `filler`, a corrupted primer separating copying from computation, and a single-factor primer separating interference from route substitution.

Conclusion

Read against the class prior rather than against zero, four of six GraphQA tasks are compromised at $n=40$: two are answered by a constant, one sits below its prior on both plain arms, and one is scored by a response bias. Within what remains, a primer that states the requested quantity acts as a second route to the answer, helping a model at floor and harming one at ceiling; added text is an active treatment whose cost is the only effect that replicates across model sizes; and the content effects that survive both controls are small and confined to two cells. For practice on these tasks and sizes, priming has near-zero expected value and turns negative as prompts lengthen, with `all` and `rwse` the worst choices on the smaller arm. For evaluation, a prompt-level graph augmentation should be reported against a class prior, a length-matched control and a graph-blind solver together; any one of them alone admits a large effect that is not graph reasoning.

Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects and no personal data. The principal cost is the GPU compute for 217,178 generations.

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